

non-rel. Schrödinger equation
or relativistic Dirac equation

LDA or GGA
or hybrids

physical orbitals or not
mesh density and basis set

$$\left(-\frac{\hbar^2}{2m} \nabla_{\vec{r}}^2 + v_{\text{ext}}(\vec{r}) + v_H(\vec{r}) + v_{xc} \right) \phi_i(\vec{r}) = E_i \phi_i(\vec{r})$$

Diagram illustrating the components of the Kohn-Sham equation and their physical interpretations:

- $-\frac{\hbar^2}{2m} \nabla_{\vec{r}}^2$: non-rel. Schrödinger equation or relativistic Dirac equation
- $v_{\text{ext}}(\vec{r})$: pseudopotential (ultrasoft/PAW/norm-conserving) or all-electron
- $v_H(\vec{r})$: Hartree potential from solving Poisson eq. or integrating charge density
- v_{xc} : LDA or GGA or hybrids
- $\phi_i(\vec{r})$: physical orbitals or not, mesh density and basis set
- E_i : view EVs as mere Lagrange multipliers or band structure approx